BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 2193 / 2198

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ. 1. 1121 — (Inter Part - I) (Session 2017-19 to 2020-22) — Mathematics (Objective) — (Group I) — Paper (I)

Q. Time Allowed:- 30 minutes — PAPER CODE 2193 — Maximum Marks:- 20

Q1. A.M between √2 and 3√2 is

(A) (B)

(C) (D)

Q2. Which of the following is an irrational number?

(A) (B)

(C) (D)

Q3. If a set S has 5 elements, Then number of improper subsets are

(A) (B)

(C) (D)

Q4. The co-factor A_{22} of the matrix [[1,2,4],[-1,2,5],[0,1,1]] is

(A) (B)

(C) (D)

Q5. The matrix [[1,2,0],[0,1,4],[0,0,6]] is

(A) (B)

(C) (D)

Q6. The quadratic equation $ax^2 + bx + c = 0$ becomes Linear equation if

(A) (B)

(C) (D)

Q7. If ω is complex roots of unity, Then value of $(3+\omega)(3+\omega^2)$ =

(A) (B)

(C) (D)

Q8. If $(7x+25)/((x+3)(x+4)) = A/(x+3) + B/(x+4)$, Then value of B is

(A) (B)

(C) (D)

Q9. G.M between 1 and 16 is/are

(A) (B)

(C) (D)

Q10. Solution of the equation cos x = -1 in [0, 2π] is

(A) (B)

(C) (D)

Q11. $(-1)^n$, n∈N is a/an

(A) (B)

(C) (D)

Q12. A die is rolled, The probability of getting 3 or an even number is

(A) (B)

(C) (D)

Q13. Middle Term (s) of $(a+b)^{11}$ is/are

(A) (B)

(C) (D)

Q14. $2\sin45° + 1/2\ \text{cosec}45°$ =

(A) (B)

(C) (D)

Q15. If $\tan\theta > 0$, $\sin\theta < 0$, Then terminal arm of the angle θ will lie in quadrant

(A) (B)

(C) (D)

Q16. If $\alpha = 30°$, then value of $\cot 3\alpha$ =

(A) (B)

(C) (D)

Q17. The period of cosec 10x is

(A) (B)

(C) (D)

Q18. If α, β and γ are the angles of an oblique Triangle, then it must be true that

(A) (B)

(C) (D)

Q19. In any Triangle ABC, with usual notations, $a/(2\sin\alpha)$ =

(A) (B)

(C) (D)

Q20. $\sin(\sin^{-1}(1/2))$ =

(A) (B)

(C) (D)

Board typos kept verbatim: 'A.M between √2 and 3√2 is', 'Which of the following is an irrational number?', 'If a set S has 5 elements, Then number of improper subsets are', 'The co-factor A_{22} of the matrix ... is', 'The matrix ... is', 'The quadratic equation $ax^2 + bx + c = 0$ becomes Linear equation if', 'If ω is complex roots of unity, Then value of $(3+\omega)(3+\omega^2)$ =', 'If $(7x+25)/((x+3)(x+4)) = A/(x+3) + B/(x+4)$, Then value of B is', 'G.M between 1 and 16 is/are', 'Solution of the equation cos x = -1 in [0, 2π] is', '$(-1)^n$, n∈N is a/an', 'A die is rolled, The probability of getting 3 or an even number is', 'Middle Term (s) of $(a+b)^{11}$ is/are', '2sin45° + 1/2 cosec45° =', 'If tanθ > 0, sinθ < 0, Then terminal arm of the angle θ will lie in quadrant', 'If α = 30°, then value of cot3α =', 'The period of cosec 10x is', 'If α, β and γ are the angles of an oblique Triangle, then it must be true that', 'In any Triangle ABC, with usual notations, a/(2sinα) =', 'sin(sin⁻¹(1/2)) =', 'improper subsets', 'co-factor'.

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QSection I. 1121 — (Inter Part - I) (Session 2017-19 to 2020-22) — Mathematics (Subjective) — (Inter Part - I) (Group I) — Paper (I)

Q. Time Allowed: 2.30 hours — Maximum Marks: 80

QSection II. Section II

Note: Attempt any three questions. (10 × 3 = 30)

Board typos kept verbatim: 'Prove that (-7/12) - (5/18) = (-21-10)/36', 'Simplify (5, -4)(-3, -2)', 'Find the multiplicative Inverse of 1 - 2i.', 'Show that the statement P → (p∨q) is tautolog' (right edge cut in scan), 'Find the inverse of the relation {(x,y) | $y^2 = 4ax$, $x \geq 0$}', 'If a, b are elements of a group G, then show that $(ab)^{-1} = b^{-1}a^{-1}$', 'Find x and y if [[x+3, 1], [-3, 3y-4]] = [[2, 1], [-3, 2]]', 'Without expansion show that |6 7 8; 3 4 5; 2 3 4| = 0', 'If A = [[1,2,-3],[0,-2,0],[-2,-2,1]], find A_{12} and A_{22}', 'Evaluate $(1 + \omega - \omega^2)^8$', 'if α, β are the roots of the equation $5x^2 - 2x + 4 = 0$, find the value of $\alpha^2 + \beta^2$', 'Show that the roots of equation $px^2 - (p-q)x - q = 0$ will be rational.', 'Write only partial Fraction Form of $(x^2+2x+3)/(x^3+x^2+1)$ without finding constants', 'Resolve $(7x+25)/((x+3)(x+4))$ into Partial Fraction.', 'If the nth term of an A.P is 3n-1 Find the A.P', 'Find the 5th term of the G.P. 3,6,12,.....', 'Find the sum of an infinite geometric series 9/4 + 3/2 + 1 + 2/3 + ...', 'If the numbers 1/k, 1/(2k+1) and 1/(4k-1) are in Harmonic Sequence, find k', 'Write (n+2)(n+1)(n) in the Factorial Form', 'How many 3-digit numbers can be Formed by using each one of the digits 2,3,5,7,9 only once?', 'If $^nC_8 = {}^nC_{12}$, find n', 'Prove the Formula 1 + 5 + 9 + ... + (4n-3) = n(2n-1) For n = 1, 2', 'Calculate $(0.97)^3$ by means of binomial theorem.', 'Expand (4-3x)^(1/2) upto 4-terms', "What is the circular measure of the angle between the hands of a watch at 4'O clock?", 'In which quadrant the terminal arms of the angle lie when sec θ < 0 and sin θ < 0', 'Prove that $\cos^2\theta - \sin^2\theta = (1-\tan^2\theta)/(1+\tan^2\theta)$', 'Find the value of tan (1110)°', 'Prove that $1 + \tan\alpha \tan(2\alpha) = \sec(2\alpha)$', 'Show that $\cot(\alpha-\beta) = (\cot\alpha \cot\beta + 1)/(\cot\beta - \cot\alpha)$', 'Find the period of cos(2x)', "Find the value of tan 19°30'", 'Find the area of the triangle ABC given three sides: a = 32.65, b = 42.81, c = 64.92', 'Find the value of r if a = 34, b = 20 and c = 42', 'Without using table/calculator Prove that $\tan^{-1}(5/12) = \sin^{-1}(5/13)$', 'Find the value of θ satisfying $2\sin^2\theta - \sin\theta = 0 : \theta \in [0, 2\pi]$', 'Find the solution of cosec θ = 2', 'Show that |x 1 1 1; 1 x 1 1; 1 1 x 1; 1 1 1 x| = $(x+3)(x-1)^3$', 'Show that the roots of $x^2 + (mx+c)^2 = a^2$ will be equal if $c^2 = a^2(1+m^2)$', 'Resolve into partial fraction $(6x^3+5x^2-7)/(2x^2-x-1)$', 'The sum of 9 terms of an A.P is 171 and its eighth term is 31. Find the series.', 'Prove that $^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$', 'Use mathematical induction to prove that the formula $1 + 1/2 + 1/4 + ... + 1/2^{n-1} = 2[1 - 1/2^n]$ is true for every positive integer n.', 'Prove that $\sin^6\theta - \cos^6\theta = (\sin^2\theta - \cos^2\theta)(1 - \sin^2\theta \cos^2\theta)$', 'Prove that $(\sin\theta + \sin3\theta + \sin5\theta + \sin7\theta)/(\cos\theta + \cos3\theta + \cos5\theta + \cos7\theta) = \tan4\theta$', 'Prove that $abc(\sin\alpha + \sin\beta + \sin\gamma) = 4\Delta s$', 'Prove that $\sin^{-1}(5/13) + \sin^{-1}(7/25) = \cos^{-1}(253/325)$', 'Answer briefly any Eight parts from the followings:-', 'Answer briefly any Nine parts from the followings:-', 'Note: Attempt any three questions.', 'multiplicative Inverse', 'tautolog'.

GROUP – II

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ. 1. 1121 — (Inter Part - I) (Session 2017-19 to 2020-22) — Mathematics (Objective) — Group II — Paper (I)

Q. Time Allowed:- 30 minutes — PAPER CODE 2198 — Maximum Marks:- 20

Q1. Partial fraction of $1/((x+1)(x^2$[illegible in source scan])) will be of the form

(A) (B)

(C) (D)

Q2. Arithmetic mean between a and b is

(A) (B)

(C) (D)

Q3. If $a_n = (-1)^n(2n-3)$ Then a_5 =

(A) (B)

(C) (D)

Q4. Multiplicative inverse of -i is

(A) (B)

(C) (D)

Q5. Tabular form of $\{x \mid x \in E \wedge 4<x<6\}$ is

(A) (B)

(C) (D)

Q6. If A = [[1,2,-2],[0,0,6],[6,7,4]] then A_{33} =

(A) (B)

(C) (D)

Q7. A matrix of order $1 \times n$ is called

(A) (B)

(C) (D)

Q8. If one root of equation $x^2 + px + q = 0$ is additive inverse of other, then

(A) (B)

(C) (D)

Q9. If ω is cube root of unity, then $\omega + \omega^2$ =

(A) (B)

(C) (D)

Q10. In any Triangle ABC, with usual notation, $(b-c)/(b+c)$ =

(A) (B)

(C) (D)

Q11. Value of $\sec(\sin^{-1}(\sqrt{3}/2)) =$

(A) (B)

(C) (D)

Q12. If Sin x = Cos x then x =

(A) (B)

(C) (D)

Q13. G.M between 2i and 8i equals

(A) (B)

(C) (D)

Q14. For independent events $P(A \cap B) =$

(A) (B)

(C) (D)

Q15. Expansion of $(1-2x)^{-1}$ is valid, if

(A) (B)

(C) (D)

Q16. $\text{Cot}^2\theta - \text{Cosec}^2\theta =$

(A) (B)

(C) (D)

Q17. $\text{Cos}(-60°) =$

(A) (B)

(C) (D)

Q18. $\text{Cos}2\alpha =$

(A) (B)

(C) (D)

Q19. Period of Cot 8x is

(A) (B)

(C) (D)

Q20. $\text{Cot}(\alpha/2) =$

(A) (B)

(C) (D)

Board typos kept verbatim: 'Partial fraction of ... will be of the form', 'Arithmetic mean between a and b is', 'If $a_n = (-1)^n(2n-3)$ Then $a_5 =$', 'Multiplicative inverse of -i is', 'Tabular form of {x | x∈E ∧ 4<x<6} is', 'If A = ... then $A_{33} =$', 'A matrix of order l×n is called', 'If one

root of equation $x^2 + px + q = 0$ is additive inverse of other, then', 'If ω is cube root of unity, then $\omega + \omega^2 =$', 'In any Triangle ABC, with usual notation, (b-c)/(b+c) =', 'Value of $\sec(\sin^{-1}(\sqrt{3}/2)) =$', 'If Sin x = Cos x then x =', 'G.M between 2i and 8i equals', 'For independent events $P(A \cap B) =$', 'Expansion of $(1-2x)^{-1}$ is valid, if', '$\cot^2\theta - \operatorname{Cosec}^2\theta =$', '$\cos(-60°) =$', '$\cos 2\alpha =$', 'Period of Cot 8x is', '$\cot(\alpha/2) =$'. CAVEAT: Q1 stem's second denominator factor and options B, C obscured by a BLANK watermark in the scan, marked [illegible in source scan].

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QSection I. 1121 — (Inter Part - I) (Session 2017-19 to 2020-22) — Mathematics (Subjective) — (Inter Part - I) Group II — Paper (I)

Q. Time Allowed: 2.30 hours — Maximum Marks: 80

QSection II. Section II

Note: Attempt any three questions. (10 × 3 = 30)

Board typos kept verbatim: 'Find the multiplicative inverse of (-4, 7)', 'Show that $\forall z_1, z_2 \in C$, $z_1 + \overline{z_2} = \bar{z}_1 + \bar{z}_2$', 'Find the difference of the complex numbers (8, 9) and (5, -6)', 'Show that the statement $(p \wedge q) \rightarrow p$ is a tautology', 'If A = {a, {b, c}}, then find P(A).', 'Write the set builder notation of the set. {0, ±1, ±2, ... ±1000}', 'Find the matrix X if: X[[5,2],[-2,1]] = [[-1,5],[12,3]]', 'Show that |a+l a a; a a+l a; a a a+l| = $l^2(3a+l)$', 'If A = [[4,λ,3],[7,3,6],[2,3,1]] is singular. Find the value of' (right edge cut in scan), 'Evaluate $(1 + \omega - \omega^2)^8$', 'Find the roots of the equation: $16x^2 + 8x + 1 = 0$ by using Quadratic formula.', 'By using remainder theorem, find the remainder when the polynomial $x^2 + 3x + 7$ is divided by' (right edge cut in scan), 'Resolve into Partial Fractions, $1/(x^2-1)$', 'Write into Partial fractions without finding the constants $9/((x+2)^2(x-1))$', 'Find the indicated term of the following sequence 1, -3, 5, -7, 9, -11, ... a_8', 'If the nth term of the A.P is 3n-1, find arithmetic progression.', 'Find the 12th term of the geometric sequence 1+i, 2i, -2+2i, -------', 'If the numbers 1/k, 1/(2k-1) and 1/(4k-1) are in harmonic sequence, find k.', 'Evaluate $^{16}P_4$.', 'In how many ways can a necklace of 8 beads of different colours be made' (right edge cut in scan), 'Find the value of n, when $^nC_5 = {}^nC_4$', 'Calculate by means of binomial theorem $(0.97)^3$', 'Expand up to 3 terms (1-x)^(1/2)', 'If x is so small that its square and higher powers be neglected, then show that $(\sqrt{(4+x)})/((1-x)^3) \approx 2 + 25/4$' (the x in 25x/4 is illegible in the scan), "Convert 54°45' into radians", 'Verify $\sin^2(\pi/6) - \sin^2(\pi/3) + \tan^2(\pi/4) = 2$', 'Prove that $\cos^4\theta - \sin^4\theta$ [operator illegible] $\cos^2\theta - \sin^2\theta \; \forall \; \theta \in R$', 'Without using tables write down the value of cos 315°', 'Prove that $\tan(45°+A)\tan(45°-A) = 1$', 'Prove that $(\sin A + \sin 2A)/(1 + \cos A + \cos 2A) = \tan A$', 'Find the period of 3 cos(x/5)', "Find the value of Cot 89°9'", 'Find the area of ΔABC having a = 200, b = 120, γ = 150°', 'In ΔABC if a = 13, b = 14, c = 15 find R', 'Show that $\sin^{-1}(-x) = -\sin^{-1}(x)$', 'Solve the equation sinx = 1/2', 'Find the solutions of sinx = -√3/2 which lie in [0, 2π]', "Use cramer's rule to solve the system of Equations", '$3x_1 + x_2 - x_3 = -4$; $x_1 + x_2 - 2x_3 = -4$; $-x_1 + 2x_2 - x_3 = 1$', 'Use synthetic division to find the values of p and q if x+1 and x-2 are the factors of the polynomial $x^3 + px^2 + qx + 6$', 'Resolve into Partial fractions $(9x-7)/((x^2+1)(x+3))$', 'If the (positive) Geometric Mean and Harmonic Mean between two numbers

are 4 and 16/5, find the numbers.', 'Prove that ⁿC_r + ⁿC_{r-1} = ⁿ⁺¹C_r', 'Find 6th term in the expansion of (x² - 3/(2x))¹⁰', 'If sinθ = -1/√2 and the terminal arm of angle is not in quad. III Find the values of remaining trigonometric functions.', 'Prove that (2sinθ sin2θ)/(cosθ + cos3θ) = tan2θ tanθ', 'Prove that r = 4R Sin(α/2) Sin(β/2) Sin(γ/2)', 'Prove that 2tan⁻¹(1/3) + tan⁻¹(1/7) = π/4', 'Answer briefly any Eight parts from the followings:-', 'Answer briefly any Nine parts from the followings:-', 'Note: Attempt any three questions.', "cramer's rule", 'Cos 2A', "Cot 89°9'", 'quad. III'.